

Trainings ▾

Preparatory Steps

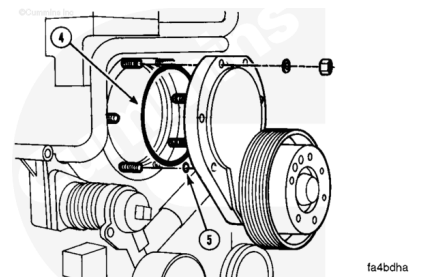
- Remove the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/18/18-008-040-tr.html>).
- Remove the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/18/18-008-002-tr.html>).



Remove

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Remove the seven fan hub mounting nuts and remove the fan hub.

Remove the o-ring (4) and seal (5).

Discard the o-ring and seal.

Inspect for Reuse

⚠ WARNING ⚠

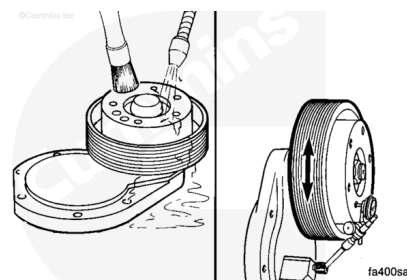
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the fan hub with solvent.

Rotate the pulley to check for rough or damaged bearings.

Check the grease seals for damage.

Check the bearing end clearance.



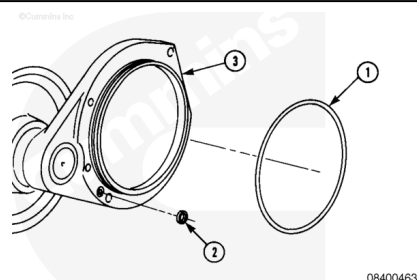
Belt Driven Fan Hub Bearing End Clearance

mm		in
0.08	MIN	0.003
0.41	MAX	0.016

If the fan hub is **not** within specifications, the fan hub **must** be reconditioned

Make sure the o-ring (1) and seal (2) are removed.

Check the support (3) for damage.



Remove the pipe plugs (4).

Install a grease fitting in one of the pipe plug holes.

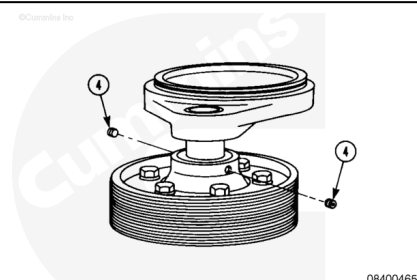
Using a grease gun, pump grease that meets the NGLI GC-LG wheel bearing grease requirements into the fan hub until it begins to come out of the open hole.

Remove the grease fitting.

Install pipe plugs (4).

Tighten the pipe plugs.

Torque Value: 15 n•m [120 in-lb]

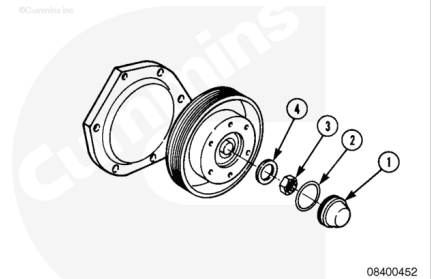


Disassemble

Remove the listed parts:

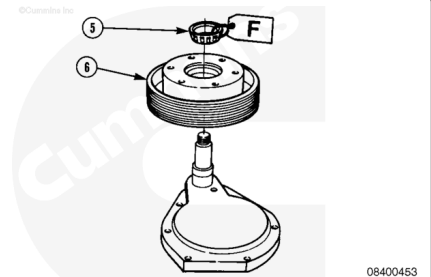
- (1) Dust cover
- (2) O-ring

- (3) Locknut
- (4) Washer.



Remove the pulley assembly (6).

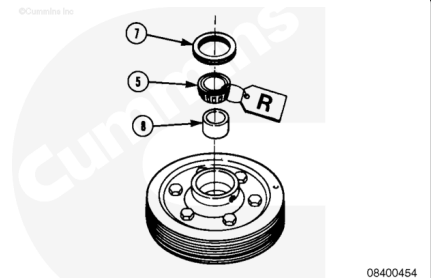
Remove and tag the front roller bearing (5).



Turn the pulley assembly over and remove the listed parts:

- (7) Seal
- (5) Inner roller bearing
- (8) Inner bearing spacer.

Note : The outer bearing spacer can not be removed until the outer bearing race is removed.



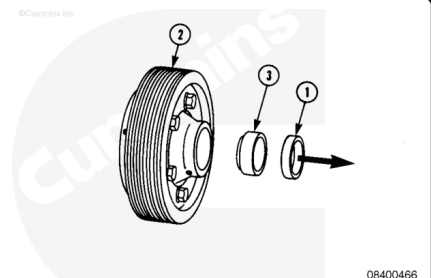
⚠ WARNING ⚠

To reduce the possibility of personal injury, wear eye and face protection when removing the bearing races. The bearing races are hardened and can chip.

Note : Do not remove the bearing race unless replacing the bearing.

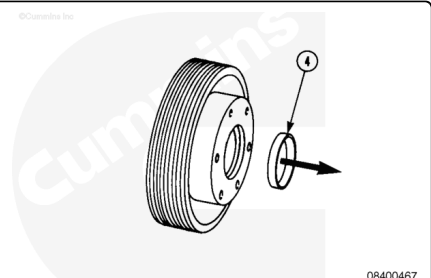
Use a hammer and a drift to drive the rear bearing race (1) out of the pulley (2).

Remove the outer bearing spacer (3).



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear eye and face protection when removing the bearing races. The bearing races are hardened and can chip.



Note : Do not remove the bearing race unless replacing the bearing.

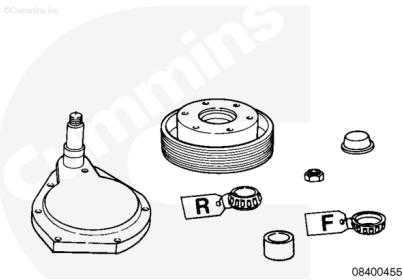
Turn the pulley over.

Use a hammer and drift to drive the front bearing race (4) out of the pulley.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not allow the force of the compressed air to turn the bearing when cleaning. Bearing damage will result because of the lack of lubrication.

Clean the parts with solvent and dry with compressed air.

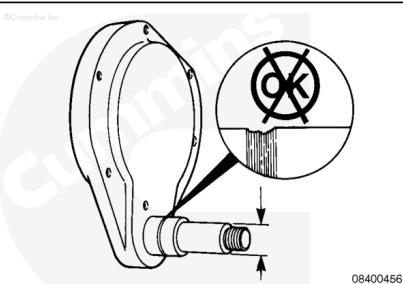
Inspect the bearings for wear.

If the bearings are worn, the bearings and bearing races **must** be replaced.

Check for shaft wear at the seal location.

If the shaft is worn, it **must** be replaced.

Measure the shaft in the bearing location.



Fan Hub Shaft Outside Diameter

mm		in
34.912	MIN	1.375

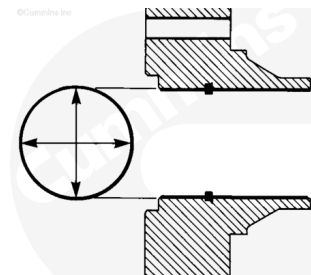
Fan Hub Shaft Outside Diameter		
mm		in
34.925	MAX	1.378

If the shaft is **not** within specifications, it **must** be replaced.

If the bearing races were removed, measure the inside diameter of the fan hub pulley at each side of the retaining ring.

Fan Hub Pulley Bore Inside Diameter		
mm		in
65.049	MIN	2.561
65.075	MAX	2.562

If the fan hub pulley is **not** within specifications, it **must** be replaced.

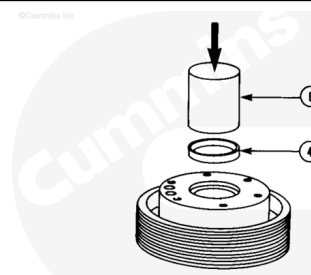


08400468

Assemble

If the front bearing race was removed, use a mandrel (6) and a arbor press to install the front bearing race (4).

Push the bearing race into the bore until it touches the retaining ring.



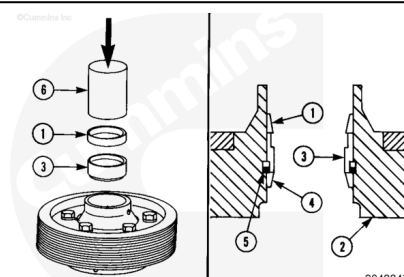
08400469

If the rear bearing race was removed, install the outer bearing spacer (3). Push the spacer until it touches the front bearing race (4).

Use a mandrel (6) and an arbor press to install the rear bearing race (1).

Push the race until it touches the spacer (3).

The spacer will **not** turn when the race is in the proper position.



08400470

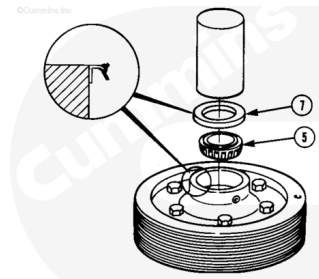
⚠ CAUTION ⚠

Do not lubricate the seal surface of the shaft. The seal and seal surface must be clean and dry.

Lubricate the bearing (5) with wheel bearing type grease that meets the NGLI GC-LG wheel bearing grease requirements and install it into the fan hub pulley.

Install the seal (7) with a mandrel.

The seal **must** be **not** more than 0.51 mm [0.020 in] below the top surface of the pulley.



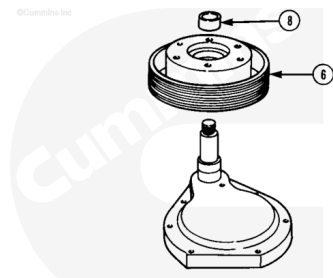
08400457

Turn the pulley (6) over and install it onto the shaft. The rear bearing **must** slide easily onto the shaft.

If it does **not** check the shaft for chips or burrs.

Install the inner bearing spacer (8).

Fill the cavity between the spacer and the pulley with water pump type grease.

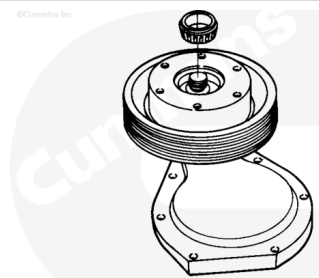


08400458

Lubricate the front bearing with grease that meets the NGLI GC-LG wheel bearing grease requirements.

Install the front bearing into the fan hub.

The bearing **must** slide easily over the shaft. If it does **not** check the shaft for chips or burrs.

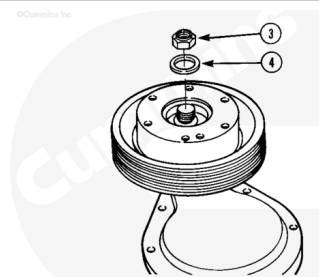


08400459

Install the washer (4) and locknut (3).

Tighten the locknut.

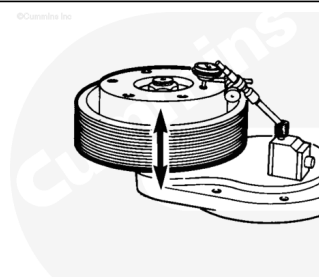
Torque Value: 205 n·m [150 ft-lb]



08400460

Measure the bearing end clearance with a dial indicator.

Belt Driven Fan Hub Bearing End Clearance		
mm		in
0.08	MIN	0.003
0.41	MAX	0.016



08400461

If the clearance is **not** within specifications, make sure the spacer has been installed correctly.

The bearings **must** touch both of the bearing spacers.

⚠ CAUTION ⚠

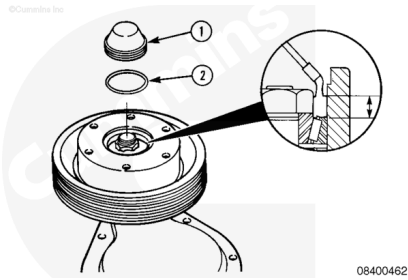
Do not use more grease than specified. Too much grease will cause overheating and failure.

Fill the cavity of the pulley to the top of the locknut with grease that meets the NGLI GC-LG wheel bearing grease requirements.

Install the o-ring (2) onto the dust cover (1).

Lubricate the o-ring with vegetable oil.

Install the dust cover.



Install

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

If the studs have been removed check the protrusion.

Stud Protrusion From Front Cover		
mm		in
50.17	MIN	1.975
51.44	MAX	2.025

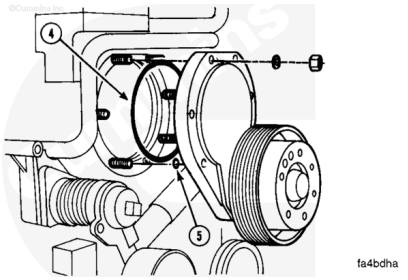
Lubricate the o-ring (4) and seal ring (5) with vegetable oil.

Install the o-ring and seal ring into the fan hub.

Install the fan hub and seven lock washers and nuts.

Tighten the nuts.

Torque Value: 35 n•m [25 ft-lb]



Finishing Steps

- Install the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/18/18-008-002-tr.html>).
- Install the cooling fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/18/18-008-040-tr.html>).



Inspect

Every 6000 hours or 2 years inspect the fan hub for wobble and grease leakage. Replace with a new or rebuilt unit as necessary.

Refer to a Cummins Authorized Repair Location for removal and replacement of fan hub.

